



BUSINESS CONNECT

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Project Title: Business Connect

Project Overview

1.Malamulele Overview

Malamulele is a developing town located in the Limpopo Province of South Africa, under the Collins Chabane Local Municipality within the Vhembe District. It plays a vital role as a commercial and service hub for nearby rural villages such as Xigalo, Mavambe, Tshikonelo, Bungeni, and others.

The town has experienced notable growth in infrastructure, housing, and small businesses, yet many local services remain underdeveloped in terms of digital presence and accessibility. Malamulele's community includes diverse individuals, from small business owners and students to professionals and visitors who travel for work or family purposes.

Despite its development, most service connections (such as finding accommodation or maintenance providers) still occur through word of mouth or physical visits, which limits business opportunities and convenience for both customers and service providers. This gap presents an opportunity for a digital platform that simplifies access to local services and accommodations.

2.Problems and Challenges

a. Lack of Online Visibility: Many local business providers (plumbers, electricians, and landlords) have no digital platform to showcase or advertise their work.

b. Manual Bookings: Guest house and room rental bookings are often made through phone calls, posters, or physical visits, which can be time-consuming and unreliable.

c. Limited Communication Channels: Customers struggle to compare business providers, check availability, or contact them easily online.

d. Inefficient Marketing: Local entrepreneurs depend mostly on word of mouth to attract clients, limiting their customer reach and growth potential.

e. Data and Availability Management: Business owners struggle to update availability (for example: which rooms are booked or vacant), leading to confusion and missed opportunities.

3. Solutions

- a. Centralized Digital Platform:** Develop a single online system where users can find and book local services such as plumbing, electrical work, room rentals, and guest houses in the Business area.
- b. Self-Managed Service Profiles:** Allow business providers and accommodation owners to create and manage their own profiles, including uploading photos, updating service details, and setting room availability.
- c. Real-Time Availability Updates:** Enable property owners to show which rooms or houses are available or occupied, ensuring accurate information for users at all times.
- d. Flexible Admin Options:** Give business owners the choice to either manage their own admin dashboard or have their details displayed and maintained by the main system administrator.
- e. Improved Accessibility and Visibility:** Promote local businesses by providing a user-friendly, mobile-accessible platform that connects customers with trusted local business providers and supports digital growth in the Business community.

Scope Statement

4. In-Scope (What the System will include)

- a. User Registration and Login:** Users (customers and business providers) will be able to create accounts to access or manage their information.
- b. Service Categories:** The system will include local services such as:
- Plumbing
 - Electrical work
 - House or room rentals
 - Guest houses, motels, and lodges
- c. Business Provider Profiles:** Each provider can manage their profile, update details, upload photos, and describe their services.
- d. Room and House Availability Management:** Property owners can indicate which rooms or houses are available or booked in real-time.
- e. Admin Panel Options:**
- Business owners can manage their own admin site, or
 - Choose to display only their details (managed by the system admin).

f. Search and Filter Functionality: Users can search by service type, location, or price range to find what they need easily.

g. Responsive Web Design: The platform will be accessible via computers, tablets, and mobile devices.

h. Booking and Contact Functionality: Users can make bookings, inquiries, or service requests through the platform.

i. Image Upload: Business providers can upload and update images of their work or rooms for display.

j. Basic Reporting and Management: Admins can view active services, bookings, and availability summaries.

5.Out of Scope (What the System will not include)

a. Online Payment Integration: The system will not include online or credit card payment processing during the first phase. Bookings will rely on manual payment on arrival.

b. Advanced Analytics or AI Features: The system will not include advanced analytics dashboards, predictive features, or AI-based recommendations at this stage.

c. Third-Party Integrations: Integration with external hotel booking systems or other third-party platforms is not included in this version.

d. Multi-Language Support: The system will operate in English only during its initial release, no other languages will be included.

Business Case

6.Business Need

Malamulele and its surrounding villages are developing rapidly, yet local business providers and accommodation owners still rely on traditional marketing methods such as word-of-mouth, posters, and physical visits. This often leads to missed opportunities, limited visibility, and poor accessibility for clients looking for trusted local services or available accommodation.

The Business Connect system addresses this gap by creating a centralized digital platform that allows residents, visitors, and businesses to easily find, book, and manage local services and rentals online. The system will promote digital inclusion, boost local economic growth, and make essential services more accessible to the community.

7.Expected benefits

- a. Digital Visibility:** Local business providers (plumbers, electricians, guest houses, etc.) gain online exposure to reach more customers.
- b. Simplified Booking Process:** Users can easily search, compare, and book services or accommodation without traveling physically.
- c. Community Empowerment:** Supports local businesses by enabling them to advertise and manage their services digitally.
- d. Increased Tourism Support:** Visitors to Malamulele and nearby areas can find verified accommodation options quickly.
- e. Operational Efficiency:** Reduces manual work and miscommunication between business providers and clients by enabling real-time updates and online visibility.

8. Business Rules

- a. User Registration:** All business providers must register using a valid email address and phone number before accessing system features.
- b. Business Provider Verification:** Only verified business providers can post or advertise their services (plumbers, electricians, landlords, guest house owners, etc.).
- c. Booking Confirmation:** Bookings are only confirmed once the business provider accepts the request or marks availability.
- d. Content Management:** Business providers are responsible for keeping their information (images, pricing, availability) accurate and up to date.
- e. Cancellation Policy:** Users can cancel bookings within a specified timeframe, and providers must update availability immediately.
- f. Admin Oversight:** System administrators have the right to remove or suspend listings that violate policies or display false information.
- g. Data Privacy:** User information will be kept confidential and used only for service and booking purposes in compliance with POPIA (Protection of Personal Information Act).
- h. Feedback and Ratings:** Customers can leave reviews and ratings, which contribute to the credibility of business providers.
- i. Payment Handling (Future Phase):** Payments will be integrated using cash or EFT upon arrival or service completion.
- j. Multi-Service Access:** A single user account can manage multiple service types.

9.Risks of the system

- a. Limited Internet Access:** Some areas in Malamulele may have poor connectivity, reducing accessibility for users.
- b. Digital Literacy:** Many local business providers may not be familiar with managing online profiles or systems, requiring training and support.
- c. Maintenance Challenges:** Regular updates and moderation are needed to keep listings accurate and prevent outdated information.
- d. Financial Constraints:** Some business providers may be unable to afford paid listings, so the system must include a free or low-cost tier.
- e. Adoption Resistance:** Locals who rely on traditional business methods may be hesitant to move online, requiring awareness campaigns to encourage participation.

10.Stakeholder Analysis

Stakeholder	Interest	Description
Project Manager	Internal	Oversees project planning, coordination, and delivery. Ensures the project stays within scope, budget, and timelines.
Business Analyst	Internal	Gathers and analyses business requirements, defines system features and design system diagrams, and ensures the solution meets stakeholder needs.
Full-Stack Developer	Internal	Designs and develops both the frontend and backend of the system, integrating all features into a functional platform.
System Tester	Internal	Responsible for testing and quality assurance to ensure the system works correctly and meets user expectations.
Project Owner & Sponsor (Maxelo Business Solutions)	Internal	The organization initiating and managing the project; provides resources, infrastructure, and oversight.
Business Owners	External	Upload and manage their services (for example: guest houses, room rentals, plumbers, electricians) on the platform to reach local customers.
Customers	External	Use the system to search, book, and connect with local services easily within Business and surrounding areas.

11.Team Roles

Team Roles	Names
Project Manager	Katlego Hope Papala
Business Analyst	Nyiko Baloi
Full-Stack Developer	Holani Baloyi
System Tester	Xitsundzuxo Nwayitelo Maluleke

Requirements Gathering

12.Functional Requirements

- a. User Registration and Login:** The system will allow both business providers and customers to create accounts and log in securely. This ensures that users can access personalized features such as managing listings or tracking bookings.
- b. Service Listing Management:** Business owners will be able to upload, edit, and delete their listings. They can include images, service descriptions, and contact details.
- c. Search and Filtering:** Customers will be able to search for services or accommodation using filters such as location, category, or price range. This helps users find relevant options quickly and easily.
- d. Booking Functionality:** The system will provide a booking module that allows customers to make reservations or service requests directly online. Confirmation messages will be sent once a booking is completed.
- e. Availability Management:** Business providers will be able to mark their rooms or services as “available” or “taken” and “unavailable” if all rooms are occupied. This ensures that customers view only up-to-date availability information.
- f. Notification System:** Users will receive automatic notifications after performing key actions, such as successful bookings, updates, or cancellations.
- g. Feedback and Rating System:** The system will allow customers to rate and review services they’ve used. This helps improve service quality and builds trust in the community.
- h. Admin Dashboard (Optional for Owners):** Business owners will have the option to manage their listings through a personal dashboard or choose to have their details displayed publicly by the system administrator.
- i. Main Admin Panel:** The system will include a central admin module for monitoring user activities, managing listings, and ensuring compliance with community standards.

j. Homepage Display and Navigation: The home page will display featured or newly added services and provide clear navigation to all main system functions.

13.Non-Functional Requirements

a. Usability: The system interface will be simple and user-friendly, enabling even users with low digital literacy to navigate and use it easily.

b. Performance: The platform will provide fast response times, with pages and search results loading within a few seconds to maintain a smooth user experience.

c. Security: The system will use secure login methods and encryption to protect user information, such as personal data and passwords.

d. Compatibility: The system will be fully functional across various devices, including smartphones, tablets, and desktop computers, to reach a wider audience.

e. Data Backup and Recovery: Regular backups will be implemented to protect data from loss due to system failure or technical issues.

f. Scalability: The system will be built in a way that allows new features, users, and service categories to be added in the future without major redesign.

g. Availability: The platform will be accessible 24/7, ensuring users can book services or check listings anytime.

h. Data Privacy: The system will comply with data protection policies to ensure that users' personal data is kept confidential and not shared with unauthorized parties.

i. Reliability: The system will operate consistently without frequent crashes or downtime, ensuring users can depend on it for accurate and up-to-date information.

j. Navigation and Visual Design: The interface have a clean layout with clear buttons, menus, and icons that guide users smoothly through different pages and actions.

14.Tools and Technologies

a. Frontend (User Interface)

Tool/Technology	Purpose
React.js	Build interactive and dynamic web pages for users (clients and service providers).
Material UI / Tailwind CSS	Create a modern, responsive, and user-friendly interface.
HTML5 / CSS3 / JavaScript	Core web technologies for building the website structure, style, and interactivity.

b. Backend (Server & Business Logic)

Tool/Technology	Purpose
Python (FastAPI)	Develop RESTful APIs to handle requests, user management, and service bookings.
Node.js (optional alternative)	Can be used for server-side development if preferred over Python.
JWT (JSON Web Token) Java, Kotlin	Secure authentication for users and service providers.

c. Database (Data Storage)

Tool/Technology	Purpose
PostgreSQL	Store all user accounts, service listings, bookings, room availability, and images.
pgAdmin	Database management tool for easy administration and queries.

d. Deployment & Hosting

Tool/Technology	Purpose
Docker	Containerize the system to ensure consistent deployment across environments.
AWS (EC2, S3, RDS) / Axxess	Cloud hosting, storage, and database services for scalability and reliability.
NGINX / Apache, VERCEL/ RENDER /RAILWAY	Web server for hosting and serving the web application.

e. Version Control & Collaboration

Tool/Technology	Purpose
GitHub	Track code versions, collaborate with the team, and manage repositories.
GitHub Projects	Manage tasks, bugs, and project progress.

f. Testing & Quality Assurance

Tool/Technology	Purpose
Jest	Frontend unit testing for React components.
Cypress	Frontend Test user interface elements like search, login, and booking forms.
Playwright	End-to-End (Full system) -Test complete booking flows — from user login → room search → booking → confirmation.
Unittest / Pytest	Backend testing for APIs, database interactions, and logic.

g. Project Management & Documentation

Tool/Technology	Purpose
Figma	Plan, assign, and track project tasks and sprints.
Draw.io	Design diagrams: ERD, Use Case, Sequence, Activity, and Precedence diagrams.
Microsoft Word	Documentation writing, reports, and versioned files.

System Design

15. Use case diagram

16. Activity Diagram

17. Context Diagram

18. Entity-Relationship Diagram (ERD)

System Specifications

19. Architecture Overview

Business Connect follows a client-server architecture with a web-based frontend and a backend server connected to a database:

a. The frontend is a responsive web application that runs in users' browsers, allowing customers and business owners to interact with the system.

b. The backend handles business logic, processes requests, manages bookings, and communicates with the database.

c. The database stores user accounts, service listings, booking information, and system logs.

d. The system is modular, allowing future expansion, such as adding new service categories or mobile app integration.

Architecture Diagram (optional for visuals):

User (Customer / Owner) → Frontend (React.js) → Backend (FastAPI) → Database (PostgreSQL)

20. Security

Security is a critical aspect to protect both users and system data:

a. Authentication: The system provides secure login and password management for customers and business owners.

b. Authorization: The system enforces different access levels (admin, business owner, customer) to prevent unauthorized actions.

c. Data Encryption: Sensitive data such as passwords and personal information is encrypted in the database.

d. Input Validation: The system validates all user inputs to prevent security vulnerabilities.

e. Regular Updates: The system is maintained and updated to patch security vulnerabilities.

21.Backup and Recovery

To prevent data loss and ensure business continuity:

a. Regular Backups: The system performs automatic backups of the database and essential files at scheduled intervals.

b. Recovery Procedures: The system provides clear processes for restoring data in case of system failure or accidental deletion.

c. Redundancy: Important data is mirrored on secondary storage for additional safety.

d. Disaster Recovery Plan: The system ensures minimal downtime in case of critical failures, maintaining system availability.

Project Plan

22.Project Schedule and Tasks

Phases	Tasks	Responsible	Duration (days)
1.Initiation & Planning	Define project scope	Project Manager	1
	Assign team roles	Project Manager	1
2.Requirements Gathering	Gather functional & non-functional requirements	Business Analyst	2
	Validate requirements with stakeholders	Business Analyst	1
3.System Design	Create Use Case diagrams, ERD, Activity diagrams, Context diagram	Business Analyst, Project Manager, System Tester &Developer	5
4.Development	Set up project environment & database	Developer	3
	Develop frontend & backend modules	Developer	13

	Integrate system features	Developer	4
5.Testing	Unit Testing	System Tester	2
	Integration & system testing	System Tester	3
6.Deployment & Training	Deploy system	Developer & Project Manager	2
	Train users & owners	Business Analyst	3
7.Maintenance & Review	Ongoing monitoring and updates	Project Manager, Business Analyst, System Tester & Developer	Ongoing

Timeline Overview

Total Duration: 40 days

If delays occur: 45 days

23.Phase timeline overview

Phase	Duration	Timeline(days)
Initiation& Planning	2	Day 1-2
Requirements Gathering	3	Day 3-5
System Design	5	Day 6-10
Development	20	Day 11-30
Testing	5	Day 31-35
Deployment& Training	5	Day 36-40
Maintenance& Review	Ongoing	Day 41+

24.Project Budget

Category	Cost elements	Estimated Cost	Notes
Deployment & Hosting	Cloud hosting for database and system	R3000	Covers server space, domain, and uptime for initial launch period
Deployment & Hosting	Deployment setup	R1000	Includes configuring server, installing software, and initial testing
Contingency Reserve	Unexpected costs	R1500	Reserved for unforeseen expenses like technical issues or extra resources
TOTAL	-	R5500	-

Testing Plan and Quality Assurance

25. Generic Pass/Fail Criteria

A test case passes if the system behaves exactly as specified, producing the correct output without errors or crashes. A test case fails if the system does not perform the expected action, produces incorrect results, or encounters errors during execution. These criteria are applied consistently to ensure all features meet the project requirements.

a. Pass

The system behaves exactly as expected.

The output matches the requirement or specification.

No errors, crashes, or incorrect results occur.

b. Fail

The system does not perform the expected action.

Outputs are incorrect, missing, or cause errors.

Crashes or freezes during execution.

26. General Approach to Testing

The general approach to testing describes how the system will be tested to ensure it meets functional and non-functional requirements. It outlines the types of testing performed, the order in which they are done, and who is responsible.

a. Unit Testing

Each module or component is tested individually (for example: login, booking, service listing).

Objective: Verify that each part of the system works as intended before integration.

b. Integration Testing

Test how modules interact with each other (for example: frontend communicates with backend, booking updates database).

Objective: Ensure different parts of the system work together correctly.

c. System Testing

Test the complete system.

Objective: Verify that all features (search, booking, notifications, dashboards) function according to requirements.

d. Usability Testing

Test with actual users (customers or business owners) to evaluate ease of use, clarity, and navigation.

Objective: Ensure the system is intuitive and user-friendly, especially for users with limited digital literacy.

e. Regression Testing

Re-test previously tested features after updates or bug fixes.

Objective: Ensure that new changes do not break existing functionality.

27.Features to be Tested

Features to be tested lists all the system functionalities that will be verified during testing. It ensures that the core and critical functions of the system work correctly before deployment.

a. User Registration and Login: Test creating new accounts, logging in, password security, and authentication.

b. Service Owner Listing Management: Test adding, updating, deleting, and displaying room or service listings with images.

c. Booking Functionality: Test that customers can book rooms or services, receive confirmations, and that bookings are saved in the database.

d. Availability Management: Test that service owners can mark rooms or services as available/unavailable, and that this updates correctly for customers.

e. Search and Filtering: Test that customers can search for services by category, location, or availability, and get accurate results.

f. Notifications: Test that customers and owners receive notifications about bookings, updates, or cancellations.

g. Feedback and Rating System: Test that customers can submit ratings and feedback, and that it displays correctly.

h. Admin Dashboard Functionalities: Test that admins can manage users, monitor listings, and oversee bookings.

i. Navigation and UI Responsiveness: Test that all screens and pages are user-friendly and responsive on different devices.

28.Features Not to Be Tested

Features Not to Be Tested lists the system functionalities that are outside the scope of the current project testing phase. This ensures that testing focuses only on critical and included features, saving time and resources.

Third-Party Integrations Not Yet Implemented

a. Future Enhancements: Features that are proposed for later versions, such as advanced analytics, additional service categories, or mobile app versions.

b. Non-Critical Background Processes: Processes that do not affect core functionality, such as logging details for analytics (if not yet implemented).

29.Test Cases

Test Case ID	Feature	Test Steps	Expected result	Pass/Fail
TC-01	User Registration	1. Navigate to registration page 2. Fill in details (name, email, password) 3. Submit form	Account is created, confirmation message is shown, login works	No result
TC-02	User Login	1. Navigate to login page 2. Enter email and password 3. Submit	User is successfully logged in and directed to dashboard	No result
TC-03	Password Reset	1. Click “Forgot Password” 2. Enter registered email 3. Submit	Password reset email is sent, user can set a new password	No result
TC-04	Service Listing Upload	1. Owner logs in 2. Navigate to “Add Service” 3. Fill in details, upload images 4. Submit	New service listing appears in the system and is visible to customers	No result
TC-05	Service Listing Update	1. Owner selects existing service 2. Update details/images 3. Save changes	Service listing updates correctly in the system	No result
TC-06	Service Listing Deletion	1. Owner selects existing service 2. Delete service	Service is removed from the system	No result

TC-07	Room Availability Management	1. Owner marks a room as available/unavailable 2. Save changes	Room availability updates correctly for customers	No result
TC-08	Search Services	1. Customer enters search criteria (category/location) 2. Submit search	Correct service listings appear matching criteria	No result
TC-09	Filter Results	1. Customer applies filters (price, availability) 2. Submit	Results reflect applied filters accurately	No result
TC-10	Booking Function	1. Customer selects service/room 2. Choose date/time 3. Submit booking	Booking is saved, confirmation message sent to customer and owner	No result
TC-11	Booking Cancellation	1. Customer navigates to bookings 2. Select a booking to cancel 3. Confirm cancellation	Booking is removed, notification sent to owner	No result
TC-12	Notifications	1. Make a booking or update service 2. Check notification system	Customer and owner receive notifications	No result
TC-13	Feedback& Rating	1. Customer submits feedback and rating for service 2. Submit	Feedback appears on the system, rating updates correctly	No result
TC-14	Admin Dashboard	1. Admin logs in 2. View all users, services, and bookings	Admin can monitor users, listings, and bookings correctly	No result
TC-15	Navigation& UI Responsiveness	1. Navigate all pages on desktop and mobile 2. Test resizing windows	All pages display correctly, buttons and links function	No result

TC-16	Logout Function	User clicks logout	User is logged out and redirected to login page	No result
TC-17	Invalid Login	1. Enter wrong email/password 2. Submit	Error message is displayed, user not logged in	No result
TC-18	Duplicate Registration	Try to register with existing email	Registration fails, proper error message shown	No result
TC-19	Image Upload Validation	Upload unsupported file type or large image	System rejects invalid file with proper error	No result
TC-20	Booking Overlap	Customer tries to book already unavailable room	System prevents booking, shows availability error	No result

30.Conclusion

The Business Connect system project aims to empower the Malamulele community by providing a unified online platform that simplifies access to local accommodation and essential home services. Through this system, residents and visitors can easily book guest houses, room rentals, and hire reliable business providers such as plumbers and electricians, all within one digital space.

The project documentation has outlined the background, objectives, requirements, design, development approach, and testing plan needed to ensure a functional and user-friendly system. By combining community needs with modern technology, the system promotes digital inclusion, supports local entrepreneurs, and contributes to the economic growth of Malamulele.

As the system moves toward implementation, continued collaboration among stakeholders, including business owners, developers, and users, will be essential for long-term success, maintenance, and scalability. Ultimately, Business Connect represents a step forward in connecting people, services, and opportunities through technology in a growing local economy.